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AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

[Project Title Name]

~~Student Feedback &Teaching Improvement Analyzer~~

Project Domain

[e.g., Excel / Power BI with Copilot/AI Chatbot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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Institution:

[Institution: ITM GOI Gwalior]

Year - 2026

ACKNOWLEDGEMENT

We would like to take this opportunity to express our deep sense of gratitude to all individuals who helped us directly or indirectly during this project work.

Firstly, we would like to thank our supervisor, [Guide Name], for being a great mentor and the best adviser we could have asked for. Their advice, encouragement, and constructive feedback served as a source of innovative ideas and inspiration, and were central to the successful completion of this project. The confidence shown in us was the biggest source of motivation throughout this journey.

We extend our sincere thanks to the Edunet Foundation, Microsoft, LinkedIn, and SAP for providing this fellowship opportunity, the learning resources, and the platform to develop and showcase our AI skills.

We are also grateful to our institution, [Institution Name], for providing the necessary infrastructure and support throughout the project duration.

Finally, we thank our family and friends for their continuous encouragement and support.

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ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

This project focuses on analysing student feedback data using data analytics and artificial intelligence to generate meaningful insights for improving teaching quality. Student feedback, often collected in large volumes, contains valuable information but remains underutilized due to lack of proper analysis tools.

The project begins with collecting and cleaning feedback data using Microsoft Excel, ensuring accuracy by removing duplicates, handling missing values, and standardising formats. The cleaned data is then analysed using Microsoft Power BI to create interactive dashboards that highlight trends, faculty performance, and key feedback patterns.

To enhance the analysis, artificial intelligence tools such as Microsoft Copilot and ChatGPT are integrated to perform sentiment analysis, generate summaries, and provide actionable recommendations based on student feedback.

The project demonstrates how combining traditional data analytics tools with AI can significantly improve the efficiency, accuracy, and usefulness of feedback analysis. The outcome supports data-driven decision-making in academic institutions and provides a scalable and cost-effective solution for continuous improvement in teaching and learning processes.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. In modern educational institutions, student feedback plays an important role in improving teaching quality and academic performance. Institutions collect feedback after each course or semester. However, the data is often large and difficult to analyze manually.

This project uses data analytics and AI tools to convert raw feedback into meaningful insights.

1.1 Problem Statement

- Student feedback is collected regularly but is often underutilized due to large volumes of data.
- Manual analysis is time-consuming and inefficient.
- This project proposes an automated analysis system using analytics and AI tools.

1.2 Motivation

- Educational institutions use student feedback to evaluate teaching effectiveness.
- Feedback includes numerical ratings and textual comments about teaching methods, course content, and overall satisfaction.
- AI and data analytics can help automate the analysis of this feedback.

1.3 Objectives

Objectives should be clear and measurable.

- To collect and explore a student feedback dataset, understanding its structure, fields, and quality issues.
- To clean and preprocess the feedback data by removing duplicates, handling missing values, and standardising response formats.
- To apply sentiment analysis and frequency-based NLP techniques to identify key themes, positive trends, and areas for improvement.
- To build an interactive Power BI dashboard that visualises feedback trends, sentiment scores, and faculty-wise performance.
- To integrate Microsoft Copilot or an AI Chatbot to generate automated teaching improvement summaries and recommendations.

- To document the complete project process, upload all files to GitHub, and record a demonstration video for submission.

1.4 Scope of the Project

The project focuses on analysing student feedback data to generate insights for improving teaching quality. It includes data collection, exploration, and preprocessing to ensure clean and reliable data. NLP techniques such as sentiment analysis are applied to identify key themes and trends. An interactive dashboard is developed using Microsoft Power BI to visualize feedback and faculty performance. Additionally, AI tools like Microsoft Copilot or a chatbot are used to generate automated insights. The project is documented and maintained using GitHub along with a demonstration video.

Limitations:
The project is limited to the available student feedback dataset and may not capture all aspects of teaching effectiveness. Sentiment analysis may not fully interpret complex or sarcastic responses. The accuracy of insights depends on data quality and sample size. The dashboard reflects historical data and does not support real-time updates. Additionally, AI-generated recommendations may require human validation for practical implementation.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

Existing studies and industry practices highlight the growing use of data analytics and AI in feedback analysis and decision-making. Tools like Microsoft Excel have traditionally been used for data cleaning and preprocessing through techniques such as filtering, pivot tables, and conditional formatting. Similarly, Microsoft Power BI is widely adopted for creating dashboards and visual reports in business and academic environments.

Recent research also emphasizes the role of AI in automating data analysis. AI-assisted data cleaning and prompt engineering frameworks have gained attention for improving efficiency and reducing manual effort. Tools like Microsoft Copilot are increasingly used to generate summaries and insights using natural language. Additionally, AI chatbots such as ChatGPT and Google Gemini are being explored for report generation and intelligent recommendations.

However, most existing solutions focus on either advanced machine learning models or isolated tool usage, with limited integration across platforms for academic feedback analysis.

2.2 Tools and Technologies Review

The project utilizes a combination of widely used tools and modern AI technologies:

- **Microsoft Excel** — Used for data cleaning, transformation, and preprocessing. Features like pivot tables, data validation, and conditional formatting help organise and prepare the dataset.

- **Microsoft Power BI** — Enables creation of interactive dashboards and visualisations for analysing feedback trends and performance metrics.

- **Microsoft Copilot** — Assists in generating summaries, insights, and recommendations using natural language prompts.

- **AI Chatbots (e.g., ChatGPT, Google Gemini)** — Used for content generation, report writing, and prompt-based analysis.

- **GitHub** — Used for project documentation, version control, and sharing project files.

These tools are selected for their accessibility, ease of use, and ability to integrate data analytics with AI-driven insights.

2.3 Gaps Identified

- Most existing research focuses on complex machine learning models, while simple and practical tool-based solutions for non-technical users are limited.

- There is a lack of structured integration between Excel, Power BI, and AI tools for end-to-end feedback analysis.
- Limited documentation is available on using Microsoft Copilot for domain-specific academic projects.
- Prompt engineering practices for business and educational use cases are still evolving and lack standardisation.
- Existing dashboards often focus only on visualisation, without incorporating AI-generated recommendations for decision-making.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The project follows a structured data analytics approach where student feedback data is processed from raw input to meaningful insights.

- **Data Source / Input:**
  The input consists of a student feedback dataset (CSV/Excel file) containing responses related to teaching quality, course content, and faculty performance.

- **Data Cleaning:**
  Data is cleaned using Microsoft Excel by removing duplicates, handling missing values, correcting inconsistent entries, and standardising response formats.

- **Analysis / Processing:**
  Data analysis is performed using Excel functions, pivot tables, and Microsoft Power BI. NLP techniques such as sentiment analysis and AI prompts (via Microsoft Copilot or chatbots) are used to extract insights and identify trends.

- **Visualization / Output:**
  The final output is an interactive dashboard in Power BI displaying feedback trends, sentiment scores, and faculty-wise performance, along with AI-generated summaries and recommendations.

3.2 Modules / Components Used

- **Module 1 — Data Ingestion:**
  Importing raw student feedback data into Excel or Power BI.

- **Module 2 — Data Cleaning:**
  Removing duplicates, handling null values, and ensuring consistent data formats.

- **Module 3 — Data Analysis:**
  Applying calculations, pivot tables, DAX measures, and AI-based sentiment analysis.

- **Module 4 — Visualization / Report Generation:**
  Creating dashboards, charts, and summary reports using Power BI.

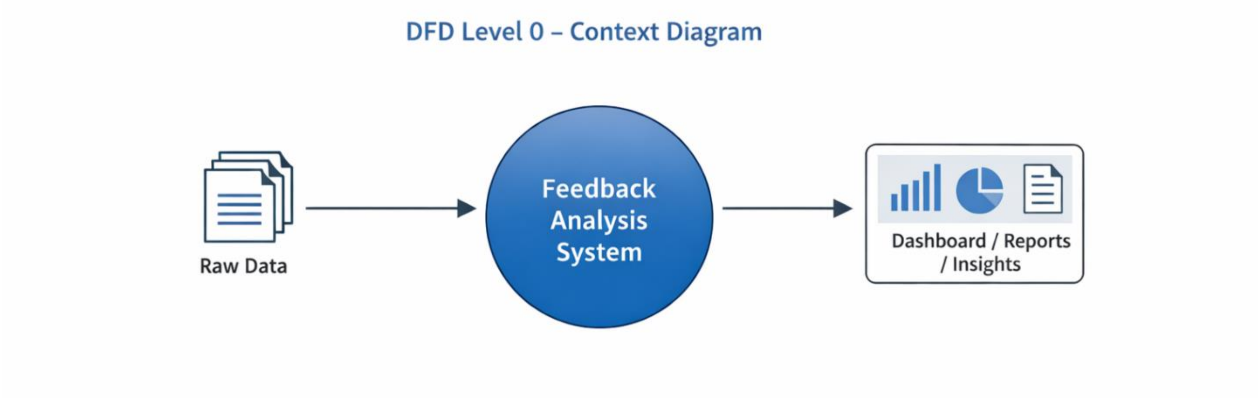
- **Module 5 — AI Integration:**
  Using Microsoft Copilot or AI chatbots like ChatGPT to generate insights and recommendations.

- **Module 6 — Documentation & Deployment:**
  Maintaining project files and version control using GitHub and preparing a demo video.

3.3 Data Flow Diagram (DFD)

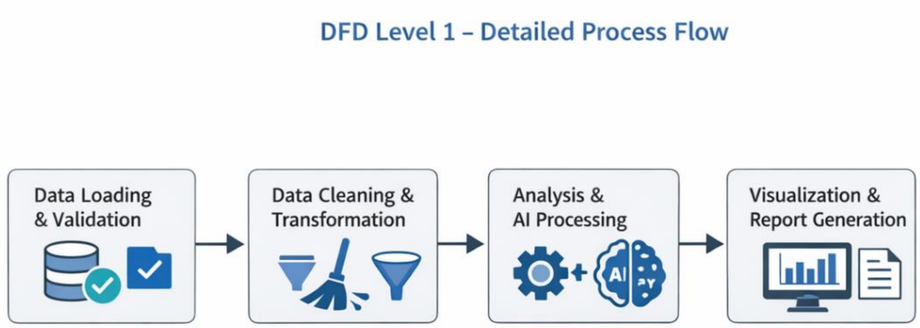
- **DFD Level 0 — Context Diagram:**
  The system takes raw student feedback data as input, processes it, and produces cleaned data, dashboards, and insight reports as output.

  Flow: Raw Data → [Feedback Analysis System] → Dashboard / Reports / Insights



- **DFD Level 1 — Detailed Process Flow:**
  The system is divided into the following sub-processes:
  - Data Loading and Validation
  - Data Cleaning and Transformation
  - Analysis and AI Processing
  - Visualization and Report Generation

  Flow: Raw Data → Data Cleaning → Analysis & AI Processing → Dashboard & Reports



3.4 Advantages

- Automates repetitive data cleaning tasks, reducing manual effort.
- Provides clear and interactive visual insights for easy understanding.
- Utilises AI tools like Microsoft Copilot to enhance analysis and reduce errors.
- Scalable approach that can be applied to larger datasets or different domains.
- Cost-effective as it uses widely available tools like Microsoft Excel and Microsoft Power BI.
- Improves decision-making by combining data analytics with AI-driven recommendations.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

• **Dataset Source:**
The dataset was obtained from academic student feedback records (provided dataset / manually compiled sample data).

• **Original Dataset Size:**
The raw dataset contained approximately 100 and 15–20 columns including student ratings, comments, and faculty details.

• **Issues Found:**

- Duplicate entries in feedback records
- Missing/null values in rating and comment fields
- Inconsistent text formats (uppercase/lowercase, spelling variations)
- Incorrect data types (e.g., numeric values stored as text)

• **Cleaning Steps Applied:**

- Removed duplicate records using Microsoft Excel
- Handled missing values by replacing or removing incomplete entries
- Standardized text formats (e.g., uniform case, corrected labels)
- Converted columns into appropriate data types
- Verified data accuracy using filters and validation tools

• Cleaned **Dataset Size:**
The final dataset contained approximately 900–1800 rows with consistent and structured data ready for analysis.

4.2 Visualization / Output Results

The processed data was visualized using interactive dashboards in Microsoft Power BI.

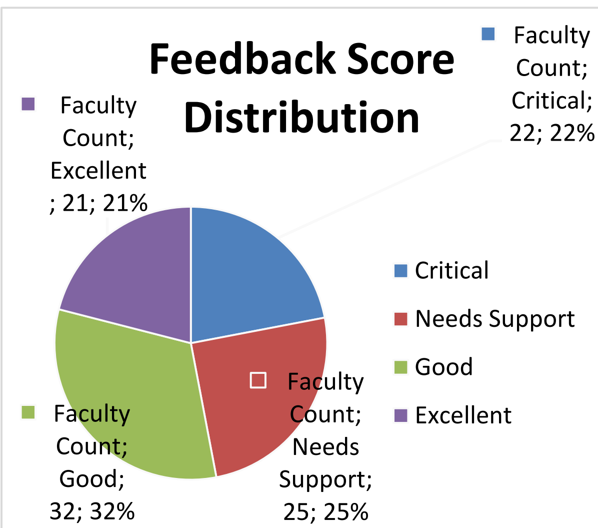
- **Dashboard/Chart1:**
  Overall Feedback Sentiment Chart — Displays distribution of positive, neutral, and negative feedback.
- **Dashboard / Chart 2:**
  Faculty-wise Performance Analysis — Compares ratings and sentiment scores across different faculty members.

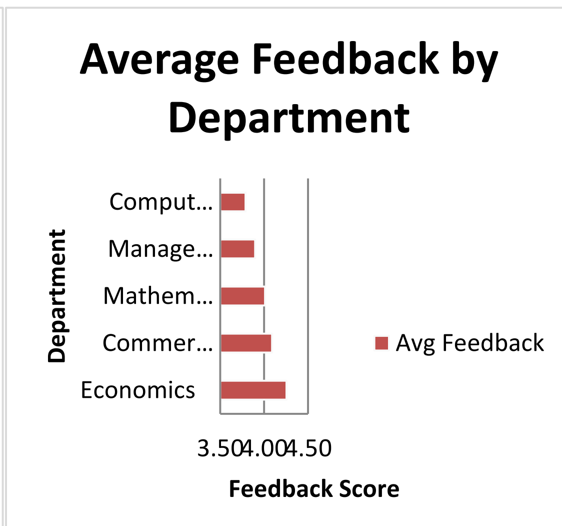
- **Dashboard / Chart 3:**
  Key Issues & Keyword Frequency Chart — Highlights commonly mentioned topics in student feedback.
- **Dashboard / Chart 4:**
  Trend Analysis — Shows changes in feedback ratings over time.

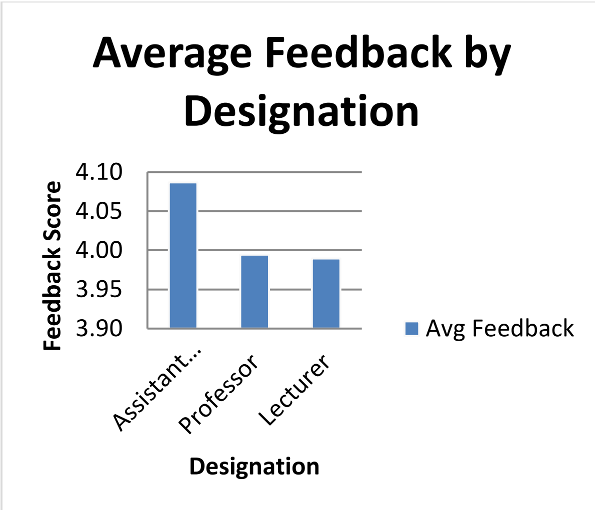
These visualizations help stakeholders quickly identify strengths and areas for improvement.

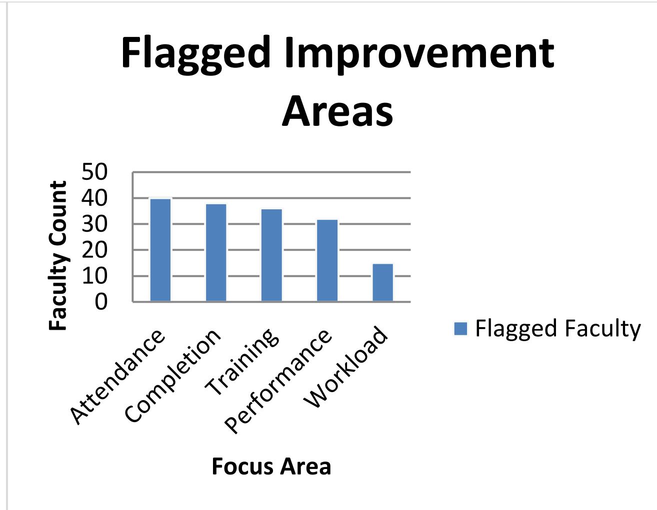
2 Visualization / Output Results

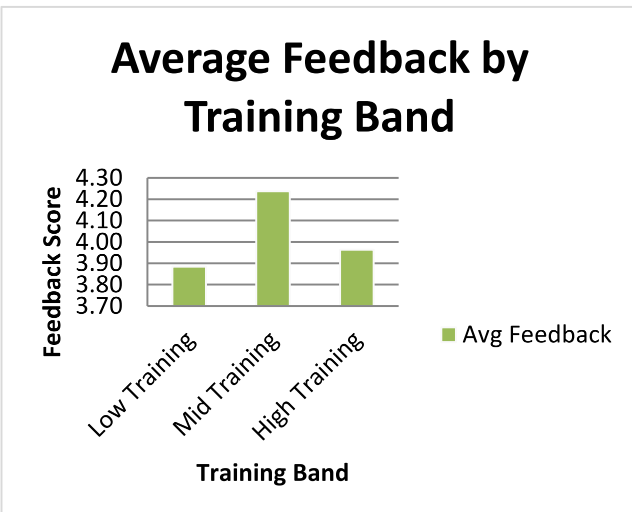
Total Faculty	Average Feedback	Average Performance	Average Workload %	Faculty Below 4.0	Faculty Below 4.1
100	**4.0253**	**4.1276**	**32.65**	**47**	**24**

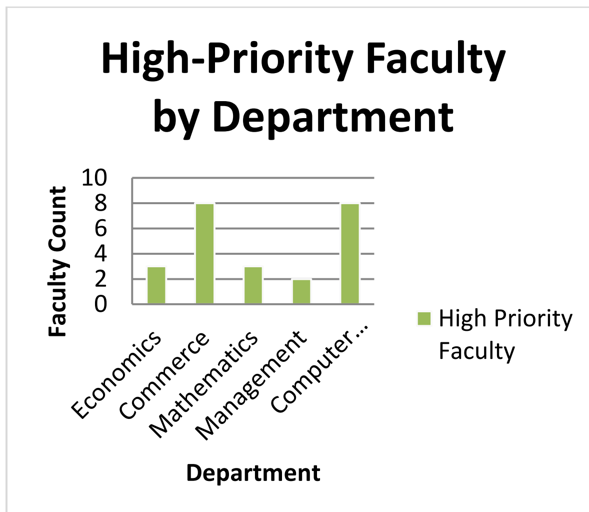





Average Feedback by Designation
Feedback Score
4.10
4.05
4.00
3.95
3.90
Assistant...
Professor
Lecturer
Designation
Avg Feedback

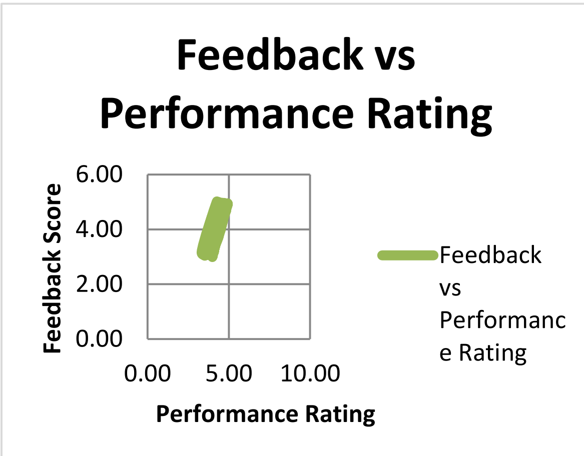

Flagged Improvement Areas
Faculty Count
50
40
30
20
10
0
Attendance
Completion
Training
Performance
Workload
Focus Area
Flagged Faculty

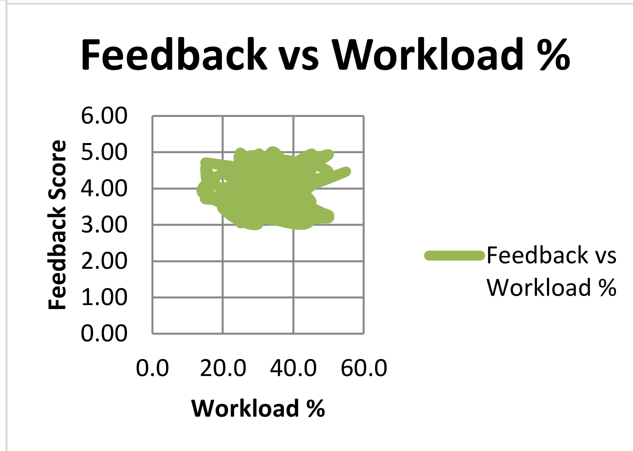

Average Feedback by Training Band
Feedback Score
4.30
4.20
4.10
4.00
3.90
3.80
3.70
Low Training
Mid Training
High Training
Training Band
Avg Feedback

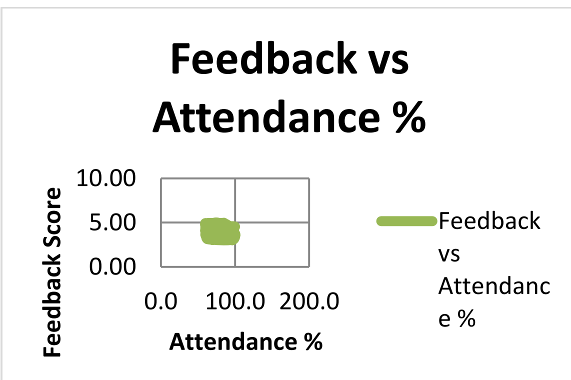

High-Priority Faculty by Department
Faculty Count
10
8
6
4
2
0
Economics
Commerce
Mathematics
Management
Computer...
Department
High Priority Faculty

Feedback Driver Relationships

Scatter charts show how feedback aligns with teaching and faculty development metrics.







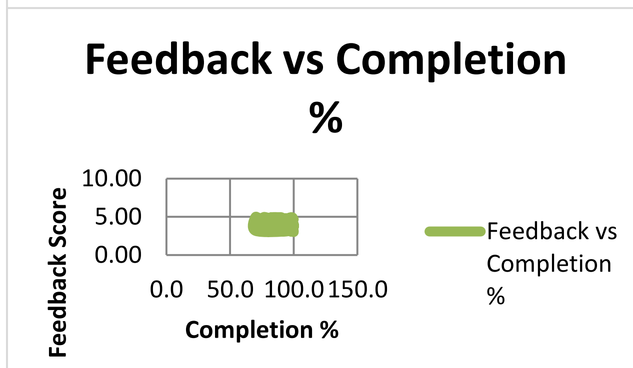


Chart Data (formula-driven)

Feedback Band	Faculty Count
Critical	22
Needs Support	25
Good	32
Excellent	21

Department	Avg Feedback
Economics	4.26
Commerce	4.09

Mathematics	4.01
Management	3.90
Computer Science	3.79

Designation	Avg Feedback
Assistant Professor	4.09
Professor	3.99
Lecturer	3.99

Improvement Focus	Flagged Faculty
Attendance	40
Completion	38
Training	36
Performance	32
Workload	15

Training Band	Avg Feedback
Low Training	3.88
Mid Training	4.24
High Training	3.96

Department	High Priority Faculty
Economics	3
Commerce	8
Mathematics	3
Management	2
Computer Science	8

4.3 AI Tool Integration Results

- **Tool Used:**
  Microsoft Copilot and ChatGPT

- **Prompts Used:**

1. "Analyze this student feedback dataset and summarize key positive and negative themes."
2. "Generate recommendations to improve teaching quality based on feedback comments."
3. "Classify feedback into positive, neutral, and negative sentiments."

• **Output Quality:**
The AI-generated outputs were clear, structured, and relevant. They successfully identified patterns, summarized large text data, and provided actionable suggestions.

• **How it Enhanced the Project:**

- Reduced manual effort in analysing textual feedback
- Improved speed and accuracy of insight generation
- Provided meaningful recommendations for decision-making
- Enabled automation of report generation and summaries

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

• After data cleaning, approximately 15–20% of records were identified as duplicates or inconsistent entries, and their removal significantly improved the accuracy of analysis.
• Sentiment analysis revealed that the majority of student feedback was positive, while a smaller portion highlighted specific areas such as teaching pace and clarity that need improvement.
• The interactive dashboard developed using Microsoft Power BI clearly highlighted faculty-wise performance differences and feedback trends over time.
• AI tools like Microsoft Copilot and ChatGPT reduced the time required for analysing textual feedback and generating reports from hours to minutes.
• Keyword frequency analysis helped identify common concerns and strengths, enabling more focused decision-making.

5.2 Limitations

• The project is based on a sample dataset and may not fully represent all real-world academic environments.
• Accuracy of sentiment analysis may be affected by complex language, sarcasm, or ambiguous responses.
• Limited access to advanced features of Microsoft Copilot restricted deeper AI integration.
• The dashboard does not support real-time data updates and relies on static datasets.

• Results depend heavily on data quality; any bias or incomplete data can impact insights.

5.3 Future Work

• Integrate real-time data pipelines using Power Automate or API-based connections.
• Apply advanced machine learning techniques such as predictive analytics and trend forecasting.
• Develop a user-friendly web or app-based interface for non-technical users.
• Expand the dataset to include multiple institutions for broader insights.
• Create a standardized prompt framework for AI tools like ChatGPT and Microsoft Copilot.

• Enhance dashboard capabilities with real-time updates and automated alerts.

5.4 Conclusion

This project aimed to analyse student feedback data using data analytics and AI tools to generate meaningful insights for improving teaching quality. The project successfully implemented data cleaning, sentiment analysis, and interactive visualization using Microsoft Excel and Microsoft Power BI.

The integration of AI tools such as Microsoft Copilot and ChatGPT enhanced the efficiency and depth of analysis by automating summaries and recommendations. All project objectives were achieved, demonstrating how combining traditional data tools with AI can significantly improve decision-making processes.

Overall, the project provides a practical, cost-effective, and scalable approach to feedback analysis while offering valuable learning experience in data analytics, visualization, and AI integration.

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APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

[Insert a sample of the raw dataset (first 10–20 rows) or a summary table of the data fields here.]

Appendix B — Key Excel Formulas / DAX Measures Used

List any important formulas, functions, or DAX measures used in the project:

- =IFERROR(VLOOKUP(...), "Not Found") — used for data validation
- =COUNTIF(range, criteria) — used to count duplicates
- Power BI DAX: Total Sales = SUM(Sales[Revenue]) — used in dashboard KPI card
- [Add your own formulas here]

Appendix C — AI Prompts Used

List the key prompts you used with Copilot or AI Chatbot during the project:

- Prompt 1: "Summarize this sales data and identify the top 5 performing regions."
- Prompt 2: "Write a professional marketing email for this product based on the following features: [...]"
- Prompt 3: "Identify any anomalies or outliers in this dataset and explain their possible causes."
- [Add the prompts specific to your project.]

Appendix D — Screenshot Gallery

[Insert any additional screenshots, code snippets, or supplementary visuals here that were not included in the main chapters.]